

FREQUENTLY ASKED QUESTIONS (FAQ)



About the Project

Q: Can I work alone or do I need a team?

A: You can work alone OR in a group of 2-3 students.

Recommendation: Groups are better because you can:

- Share computational resources
- Divide tasks (data collection, model training, demo creation)
- Learn from each other
- Tackle more ambitious projects

But working alone is perfectly fine if you prefer!

Q: What tier should I choose?

A:

- **First CV project?** → Tier 1 (safe, achievable)
- **Done ML before?** → Consider Tier 2
- **Very experienced?** → Tier 3

Remember: Tier 1 done excellently = 100%! Bonus points are BONUS.

Q: Can I change my project idea after the midterm?

A:

- **Minor changes:** Yes, totally fine!
 - **Major changes:** Need instructor approval
 - **Best practice:** Choose a solid idea for midterm that you can iterate on
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Q: What if my project is too ambitious?

A: Scale it down! It's better to:

- Complete a simple project successfully
- Than fail at a complex one

Talk to instructor ASAP if you're worried.

Q: Can I use existing code/libraries?

A: Yes!

Encouraged:

-  Use Ultralytics YOLO
-  Use Hugging Face models
-  Use PyTorch/TensorFlow
-  Use OpenCV

With attribution:

-  Use code from tutorials (cite source)
-  Use Stack Overflow solutions (cite)
-  Modify GitHub examples (cite)

Not allowed:

-  Submit someone else's complete project
-  Copy without understanding
-  Use code without attribution



About Resources & Compute

Q: I don't have a GPU. Can I still do this project?

A: Yes! You have several free options:

1. **Google Colab** (free GPU, ~12 hours/day)
2. **Kaggle Notebooks** (free GPU, 30 hours/week)
3. **Heidi Platform** (ask instructor for access)

Tier 1 projects work perfectly on Colab free tier!

Q: What if Google Colab keeps disconnecting?

A: This is normal! Solutions:

1. Save your work frequently (Ctrl+S)
2. Save model checkpoints to Google Drive
3. Use Kaggle (longer sessions)
4. Work in shorter sessions

Q: Do I need to pay for anything?

A:

- **Tier 1 projects:** 0(*completely free!*) — * * *Tier2 projects* : * * *Maybe* 5-10 if using paid APIs
- **Tier 3 projects:** Up to \$10 for APIs

You don't need to spend money to get a good grade!

Q: What's the Heidi Platform?

A: Adaptive Computing cluster with A100 GPUs.

- Ask instructor for access
 - Good for Tier 2/3 projects
 - More powerful than Colab
 - Requires learning cluster computing
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Q: Can I use my own computer?

A: Yes, if you have:

- NVIDIA GPU with 6GB+ VRAM
- 16GB+ RAM
- And you know how to set up CUDA

But most students use Colab - it's easier!

About Teams

Q: How do I find teammates?

A:

1. Ask in class
2. Post on Canvas discussion board
3. Form study groups
4. Talk to people sitting near you

Tip: Find people with complementary skills!

Q: What if my teammate isn't contributing?

A:

1. Talk to them first (maybe they're struggling?)
2. Document who does what in README

3. Contact instructor if serious issues
4. Each person documents their own AI usage

We can assess individual contributions!

Q: Can I switch teams after midterm?

A: Generally no, but talk to instructor if there's a serious issue.



About Deliverables

Q: Do I need to write a long report?

A: No!

What you need:

-  Good README (documentation)
-  Code that works
-  Demo video
-  Presentation slides

What you don't need:

-  Long written report
-  Essay about computer vision
-  Literature review

Focus on building, not writing!

Q: What if my demo fails during presentation?

A:

1. **Don't panic!**
2. Play your backup video
3. Explain what went wrong
4. Answer questions anyway

We understand tech demos fail sometimes. Having a backup shows professionalism!

Q: How long should my demo video be?

A: 3-5 minutes

Structure:

- 30 sec intro
 - 2-3 min live demo
 - 1 min results
 - 30 sec conclusion
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Q: Do I need to deploy my system?

A: No!

What's required:

- ☒ Works in Jupyter notebook
- ☒ Can demo it live

What's optional (bonus):

- Nice to have: Streamlit/Gradio interface
 - Not needed: Full web deployment
 - Not needed: Mobile app
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About Grading

Q: Do I need perfect accuracy to get a good grade?

A: No!

We grade on:

- Does it work? (most important!)
- Do you understand it?
- Is it well-documented?
- Did you learn from it?

Good analysis of 70% accuracy > no analysis of 95% accuracy

Q: What if I don't finish on time?

A:

1. Submit what you have! Partial credit for partial work
 2. Contact instructor BEFORE deadline if struggling
 3. Better to submit something incomplete than nothing
 4. We can work with you if you communicate early
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Q: Can I get 100% with a Tier 1 project?

A: Yes!

- Tier 1 done excellently = 100%
- Tier 2 = up to 110%
- Tier 3 = up to 120%

Bonus points are BONUS - not required for full credit!

Q: How important is the live demo?

A: Very important! It's 5 minutes of your 10-minute presentation.

Shows:

- Your system actually works
 - You understand it
 - You can debug issues
 - You're not just showing slides
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About AI Usage

Q: Can I use ChatGPT/Claude/Copilot?

A: Yes! But you must:

1. Use it to LEARN, not to avoid learning
2. Understand all AI-generated code
3. Document usage in AI log
4. Be able to explain everything

See full AI Usage Policy document for details.

Q: How much AI usage is too much?

A:

- Too much: "Build my entire project"
- Good: "Explain how YOLO works"
- Good: "Debug this error"
- Good: "Generate HTML for my demo"

If you can't answer questions about your code, that's too much!

Q: Do I really need to document AI usage?

A: Yes! It's required.

Shows:

- Honesty about what you used AI for
- What you learned
- You used AI to learn, not to cheat

5-10 entries minimum in `docs/AI_usage_log.md`



About Data

Q: Where do I find datasets?

A: Check these sources:

1. **Roboflow Universe** - 100,000+ CV datasets
 2. **Kaggle Datasets** - Great for competition data
 3. **Hugging Face** - ML datasets
 4. **Google Dataset Search** - Search engine for datasets
 5. **Collect your own** - Phone camera!
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Q: Can I collect my own data?

A: Yes!

Easy ways:

- Take photos with your phone
- Download from Google Images (be careful of copyright!)
- Screen capture from videos
- Generate synthetic data

Important: Need 100+ images minimum for training

Q: What if I can't find the right dataset?

A:

1. Search harder! Try different keywords
 2. Combine multiple datasets
 3. Collect your own (easier than you think!)
 4. Change your project idea
 5. Ask instructor for suggestions
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Q: Do I need to label my data?

A: Depends:

- **Using public dataset?** Already labeled!
- **Collecting own data?** Yes, you'll need to label

Labeling tools:

- Roboflow (free, easy)
 - Labellmg (for object detection)
 - Label Studio (advanced)
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Technical Questions

Q: Which model should I use?

A: For beginners:

- **Object Detection:** YOLOv8 (easiest to use!)
- **Classification:** ResNet50 (standard baseline)
- **VLM:** BLIP (good balance)

More important than which model = understanding how to use it!

Q: Python is my only programming language. Is that OK?

A: Yes! Perfectly fine!

If you need other languages (JavaScript, HTML, etc.), use AI to generate them. Focus your learning on Python and CV concepts.

Q: I'm getting errors. What should I do?

A:

1. Read the error message carefully
2. Google the error
3. Ask AI (ChatGPT/Claude)
4. Check Stack Overflow
5. Ask classmates on Canvas
6. Ask instructor

Debugging is a skill - you'll learn a lot from fixing errors!

Q: How do I know if my model is good?

A: For object detection:

- $mAP50 > 0.5$ = okay
- $mAP50 > 0.7$ = good
- $mAP50 > 0.9$ = excellent

For classification:

- Accuracy $> 70\%$ = okay
- Accuracy $> 85\%$ = good
- Accuracy $> 95\%$ = excellent

But context matters! Some tasks are harder than others.

Q: My model is overfitting. What should I do?

A:

1. Get more training data
 2. Use data augmentation
 3. Add dropout layers
 4. Use early stopping
 5. Reduce model complexity
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Q: Training is taking too long!

A:

1. Use smaller model (yolov8n instead of yolov8x)
 2. Reduce image size ($640 \rightarrow 320$)
 3. Use fewer epochs
 4. Check if GPU is being used
 5. Use better GPU (Kaggle or Heidi)
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About Specific Project Types

Q: Is object detection easier than classification?

A: For this course, object detection (with YOLO) is actually easier because:

- Ultralytics YOLO is very user-friendly
- Tons of tutorials available
- Works well out-of-the-box

Classification requires more fine-tuning knowledge.

Q: Should I do a VLM project?

A: Only if:

- You're comfortable with Tier 2 complexity
- You're interested in multimodal AI
- You have access to APIs (free or paid)
- You want the bonus points

Not required for a good grade!

Q: Can I do a 3D vision project?

A: Yes, but:

- This is Tier 3 territory
- Requires more compute
- More complex than 2D
- Make sure you have time

Talk to instructor first!



About Timeline

Q: Can I start late?

A: Not recommended!

CV projects take time:

- Data collection: 1-2 weeks
- Training: 1-2 weeks
- Testing & debugging: 1 week
- Demo creation: 1 week

Start early, iterate often!

Q: What if I finish early?

A: Great! You can:

1. Improve accuracy
2. Add features (move to higher tier)
3. Make better documentation
4. Help classmates (bonus points!)

5. Start early on final exam prep

Q: Is there a deadline extension?

A: Contact instructor BEFORE deadline if you need one.

Valid reasons:

- Medical emergency
- Family emergency
- Technical disasters

Not valid:

- "I forgot"
 - "I was busy with other classes"
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Learning & Development

Q: I've never done CV before. Can I succeed?

A: Yes! This course is designed for beginners.

Follow:

1. Getting Started Guide
2. Choose Tier 1 project
3. Use provided code templates
4. Ask for help when stuck
5. Start early

Many students with no CV experience do great!

Q: What if I don't understand something?

A:

1. **Ask!** No question is dumb
2. Use AI to learn (ChatGPT/Claude)
3. Watch YouTube tutorials
4. Read documentation
5. Ask classmates
6. Ask instructor

Learning is the goal, not knowing everything already!

Q: Will this help me get a job?

A: Yes! Computer vision is in high demand.

This project gives you:

-  Portfolio piece on GitHub
 -  Hands-on experience
 -  Something to talk about in interviews
 -  Understanding of production ML
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Miscellaneous

Q: Can I use this project for another class?

A: Check with both instructors first!

Generally:

- Different projects for different classes
 - Can't submit same work twice
 - But can build on previous work
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Q: What if my idea is similar to someone else's?

A: That's okay!

- Implementation will be different
- Your data will be different
- Your results will be different

Just don't copy their code!

Q: Can I continue this project after the course?

A: Yes! Please do!

Many students:

- Improve it for their portfolio
- Turn it into a startup
- Publish it as open source
- Use it for research

That's the best outcome!

Q: Where can I get help?

A:

1. **Canvas Discussion Board** - Ask classmates
2. **Office Hours** - Ask instructor
3. **Email** - For private questions
4. **AI Assistants** - For technical help
5. **Documentation** - Read the docs!

Don't suffer in silence - ask for help!



Emergency Questions

Q: I'm completely stuck and don't know what to do!

A:

1. **Don't panic!**
2. Email instructor ASAP
3. Come to office hours
4. Break problem into smaller pieces
5. Focus on getting SOMETHING working

We're here to help you succeed!

Q: My laptop died / GPU crashed / lost all my work!

A: Prevention:

- ☒ Use GitHub (auto-backup!)
- ☒ Save to Google Drive
- ☒ Keep model checkpoints

If disaster happens:

- Email instructor immediately
 - We'll work something out
 - Learn from it
-

Q: I'm overwhelmed. Is this project too much?

A:

1. Talk to instructor

2. Scale down to simpler project
3. Focus on Tier 1
4. Break it into small steps
5. Take it one day at a time

You CAN do this! 🚀

💡 Still Have Questions?

Ask the instructor!

- In class
- Office hours
- Canvas message
- Email

No question is too basic. We want you to succeed!

Start coding or [generate](#) with AI.